Supply Chain & Inventory Analytics

Portfolio project — final analysis pack

This project examines demand, inventory health, reorder-threshold exposure, sales performance, and supplier/warehouse patterns using the public dataset bundled in the original repository.

Dataset scope

Metric	Value
Observations	91,250
SKUs	50
Warehouses	5
Suppliers	10
Regions	4
Date range	2024-01-01 to 2024-12-30

Core results

KPI	Result
Units sold	1,829,979
Sales value*	33,426,337.22
COGS*	22,338,135.99
Gross profit*	11,088,201.23
Gross margin	33.17%
Below reorder-point observations	4,787 (5.25%)
Stockout-flag observations	0

Analytical approach

• Calculated sales value as Units_Sold × Unit_Price, cost of goods sold as Units_Sold × Unit_Cost, and gross profit as sales value less COGS.

• Defined reorder exposure as Inventory_Level < Reorder_Point. The reported 5.25% is a proportion of daily records, not a percentage of unique SKUs.

• Compared inventory coverage proxy (Inventory_Level / Units_Sold for the row) with supplier lead time. This is a rough observation-level proxy, not a demand forecast.

• Prepared SQL KPI queries, quality checks, and aggregate views for SKU, warehouse, and supplier analysis.

• Created a browser dashboard that shows monthly demand, top SKUs by sales value, warehouse reorder risk, and supplier lead time.

Interpretation and limitations

The dataset does not specify currency, so financial values are presented in generic monetary units. Inventory_Level appears in daily observations; summing its value over all dates may double-count recurring stock snapshots and should not be interpreted as a point-in-time inventory balance. Reorder threshold exposure signals records that merit review, but

does not by itself prove an actual or future stockout. The analysis uses the source dataset as supplied and preserves the source project attribution.

Source and attribution

Original repository and dataset: github.com/HarshilNarola/supply-chain-inventory-analytics. This portfolio pack is a supplementary analysis and presentation of the published dataset; it does not claim authorship of the original dataset or repository.